PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS

39th ANNUAL CONVENTION

Fueling Progress: Shaping Animal Nutrition for a Competitive and Sustainable Future

6 OCTOBER 2026

Grand Ballroom, Okada Manila










PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS
39th Annual Convention | October 6, 2026

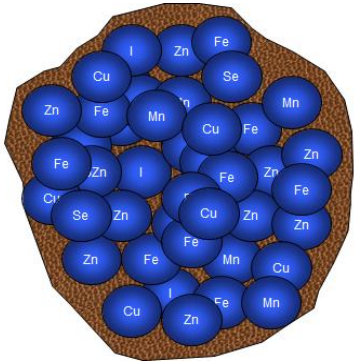
Encapsulating All Trace Minerals for Precision, Simplicity and Performance

Kevin Liu, Ph.D.
Syno Int. P/L, Singapore

Encapsulating All Trace Minerals for Precision, Simplicity and Performance



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Outline

1. Why trace minerals?
   - ✓ Cu, Fe, Mn, Zn, I, Se and Co are essential nutrients
   - ✓ Serious negative impacts to feed and gut health
2. Evolution
   - ✓ 1st and 2nd generations (ITM, OTM)
3. Metal Polysaccharides Complex (MSPC)
   - ✓ Encapsulating all trace minerals, “All-in-One”, the 3rd generation
   - ✓ Precision, simplicity, performance, sustainability

ITM & OTM: Inorganic and Organic Trace Minerals

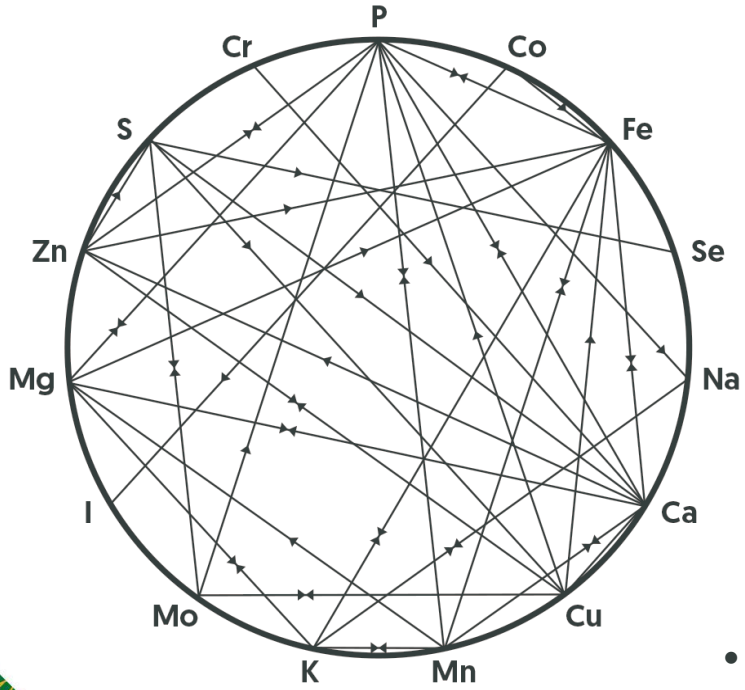
1. Essential with cautions

Discovering side effects



- ✓ **Trace minerals Fe, Cu, Mn, Zn, Se and I are essential nutrients**
- ✓ **They are reactive and corrosive** (*see the hands of farm operator, mixing feeds with water using bare hands for a few weeks*)
- ✓ **When used in premix and feed, the minerals salts are destructive to other components and GI tract**

Minerals: Reaction & Antagonism



- There are > 30 reactions among minerals used as feed additives
- As production challenges intensify, fine-toning for a balanced Macro- and Micro-mineral nutrition is important

- Mineral wheel of absorption concept

How much trace minerals we use?

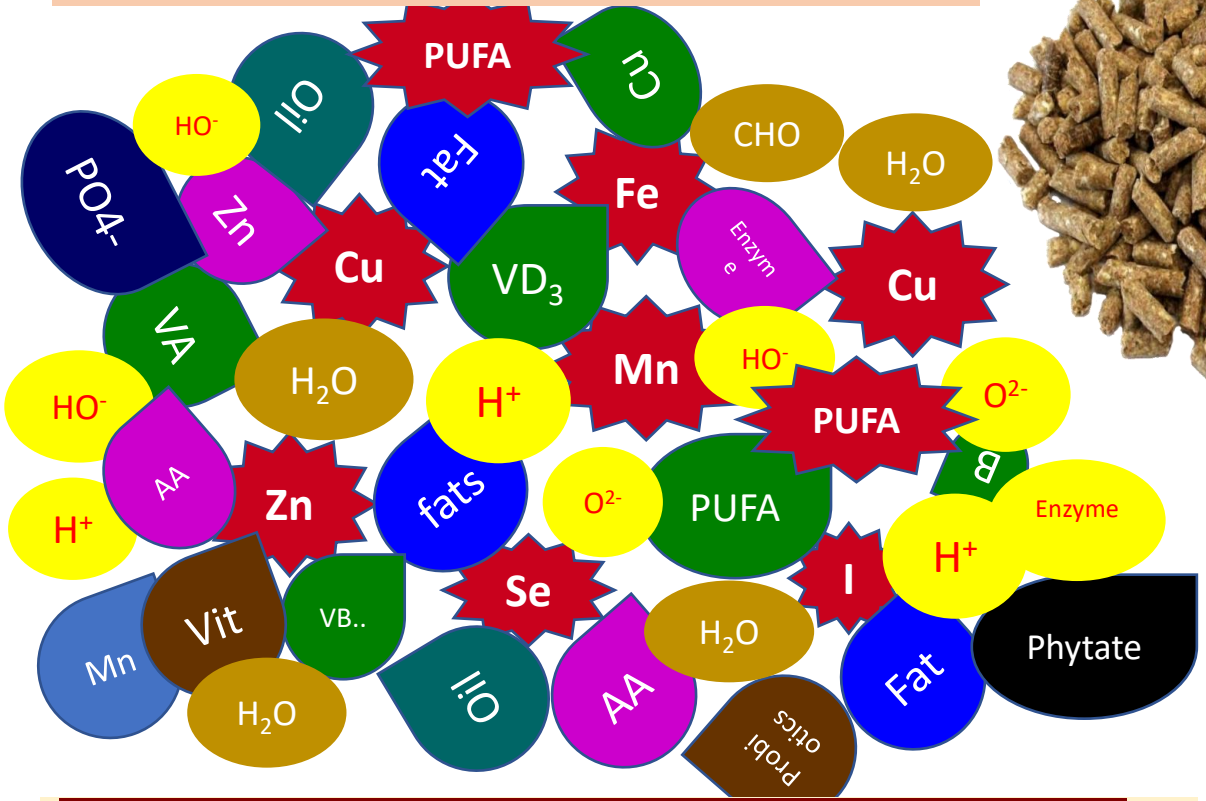
mg/kg diets	Molecule	Total g/mt	SO_4^{+2}, %	SO_4^{+2} g/mt
Cu, 15	$CuSO_4 5H_2O$	60	38.47	23.08
Fe, 40	$FeSO_4 H_2O$	129	56.53	72.92
Mn, 80	$MnSO_4$	312.5	63.6	198.75
Zn, 100	$ZnSO_4 H_2O$	220	53.6	117.8
Total, g/mt		**721.5**		**412.55**

- ✓ We add metal ions, with > 412 g strong acid SO_4^{-2}/mt diet, plus PO_4^{-3} etc.,
- ✓ The chemicals release ions that react and re-bond with other molecules rapidly.
- ✓ Strong acids, oxidative, bad taste and reduce feed palatability.

Cu, Fe, Mn, Zn catalyse oxidation

Feed is a mixture





Oxidations deteriorate feed quality

Feeds: Fresh vs. Aged

Change in FCR of a branded feed after different storage times Oct. 2001						
Days of Feed Storage	**Pig (55~135 Days)**			**Broiler**		
	ADG, g	FCR	Mortality %	Live wt on Day 43, g	FCR 1-43 D	Mortality %
1-5 D	**847**	**2.53**	**0.46**	**2560**	**1.62**	**1.35**
40-45 D	**722**	**2.87**	**1.02**	**2310**	**1.89**	**3.71**

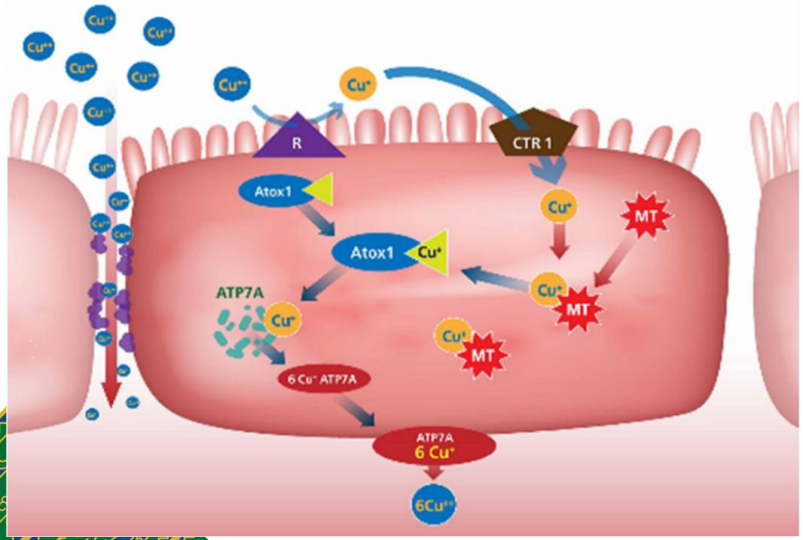
Note: Compare FCR of a branded feed, marked shelf-life 45 days, at Day 5 and near its end of shelf-life. **Lab analyses show PUFA reduced by approx. 75%; Vit A and D decreased by approx. 87%.**



Broiler Starter, Pullets

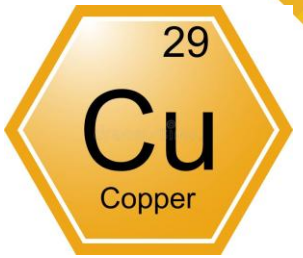
Wet Dropping

- ✓ Oxidative damages in GI Tract
- ✓ Trace minerals encapsulation solved the problem





Regulation changes - Cu, Zn



Cu: A reduction from 170 mg to 25 mg/kg feed in piglets would save 1200 ton Cu/year being spread in the field, and reduce total Cu emission from farm animal production by about 20% (FAADAP)

Zn: concerns on heavy metal accumulation in the environment and the potential for antimicrobial resistance, the max. level of Zn permitted in animal feed will reduce to 150 mg/kg in June 2022, high levels of ZnO will no longer be permitted (EU).



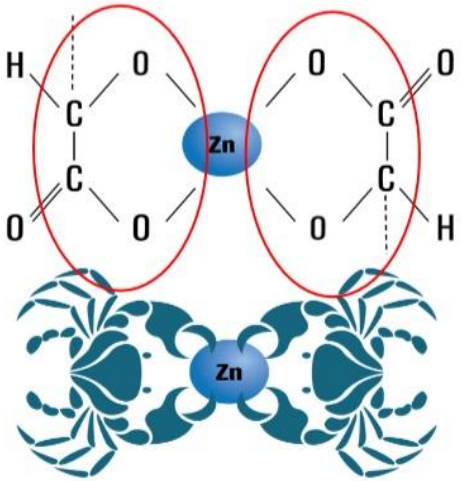
One important strategy to reduce trace minerals in diets to mitigate the environmental issue, without affecting animal performance, is inclusion of OTM which have greater bioavailability.

Over the years, research has progressed from using solely OTM to partially replacing ITM, to total replacement, with no negative effect on performance parameters. EFSA, 2016, 14, 4563.

2. Organic Trace Minerals

Advantages

1. Covalent bindings with organic ligands, metal complex (polysaccharides, amino acids, proteinates, organic acids), etc.
2. More stable, reduced solubility and reactivity, higher absorption and retention
3. Lower dosage, less excretion, environmental benefits
4. Partial or full replacement of ITM by OTM can support expected growth and health



(Kratzer F.H., & Vohra P. 1986. Chelates in Nutrition. CRC Press Inc.)

Byrne & Murphy, 2022

Organic Zn

1. Variable in composition / solubility
2. Generally equal to Zinc Sulphate

Chemical characteristics and relative bioavailability of supplemental organic zinc sources for poultry and ruminants[1,2]

J. Cao*,[3], P. R. Henry*, R. Guo*,[4], R. A. Holwerda‡, J. P. Toth†,[5], R. C. Littell§, R. D. Miles¶, and C. B. Ammerman*,[6]

Departments of *Animal Science, †Food Science and Human Nutrition, §Statistics, and ¶Dairy/Poultry Sciences, University of Florida, Gainesville 32611-0900 and ‡Department of Chemistry & Biochemistry, Texas Tech University, Lubbock 79409

Table 2. Chemical composition of zinc sources on an as-fed basis[a]

Source	DM, %	N, %	Zn, %	Ca, %	P, %	Mg, %	Cu, mg/kg	Fe, mg/kg	Mn, mg/kg
Zn Methionine A	90.7	.97	3.78	2.99	.027	.10	13.3	5,456	158
Zn Methionine B	91.3	1.81	9.29	3.20	.009	.05	32.0	12,166	181
Zn Polysaccharide	87.6	.45	19.02	.61	.079	.25	31.4	334	37
Zn Lysine	91.7	3.10	9.43	2.30	.009	.07	39.0	747	608
Zn Amino acid chelate	86.0	5.79	9.42	.17	.318	.22	144.7	191	312
Zn Proteinate A	93.7	7.19	13.63	.28	.262	.14	93.3	378	962
Zn Proteinate B	79.7	1.79	13.65	.16	.454	.22	28.2	752	339
Zn Proteinate C	83.8	4.14	13.01	.17	.277	.09	6.5	1,033	132
Zn Sulfate	62.7	ND[b]	20.8	ND	ND	ND	.7	ND	.5

[a]Each value based on triplicate determinations except for N content which had duplicate determinations.
[b]Not detectable.

J. Anim. Sci. 2000. 78:2039–2054

Organic Se

High proportion of inorganic Se

Inorganic and Organic Selenium Speciation of Seleno-Yeasts Used as Feed Additives: New Insights from Elemental Selenium Determination

Mohammed A. Hachemi[1] · Denise Cardoso[1] · Michele De Marco[1] · Pierre-André Geraert[1] · Mickael Briens[1]

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Abstract
Seleno-Yeasts (SY) used as feed additives are known to contain different Selenium (Se) species. Seleno-Yeasts has been shown, on previous analytical methods, to contain selenomethionine (SeMet), selenocysteine (SeCys), selenate (Se^{IV}) and selenite (Se^{VI}), and various other organic and inorganic Se forms identified but rarely quantified. A new advanced method has allowed elemental Se (Se^0), an inorganic Se species, to be quantified, thereby obtaining better insight into the proportion of inorganic Se in SY products. The study aimed to quantify the Se^0 in SY products and assess the proportion of inorganic Se in SY. The Se speciation of 13 fresh commercials SY from different suppliers and batches, was assayed for the total Se, inorganic Se species (Se^{IV}, Se^{VI} and Se^0), and organic Se species (SeMet and SeCys). Results on total Se were in line with the expected Se concentrations for all evaluated samples. The proportion of Se present as Se^0 ranged from 3.6% to 51.8%. The quantity of Se^0 in the SY products, added to Se^{IV} and Se^{VI}, indicated an average proportion of inorganic Se of 14.2% for the 13 analyzed SY products. The proportion of Se as SeMet ranged from 19.0% to 71.8%, (average of 55.8%), and a large variability in the SeMet content was observed. The SeCys content was also variable, with an average of 3.8%, relative to the total Se. In conclusion, advances in the analytical characterization have revealed that SY products can have a significantly high proportion of inorganic Se, which could affect the bioavailability of Se from SY supplements and explain their variable and lower bio-efficacy than pure SeMet supplements, such as hydroxy-selenomethionine.

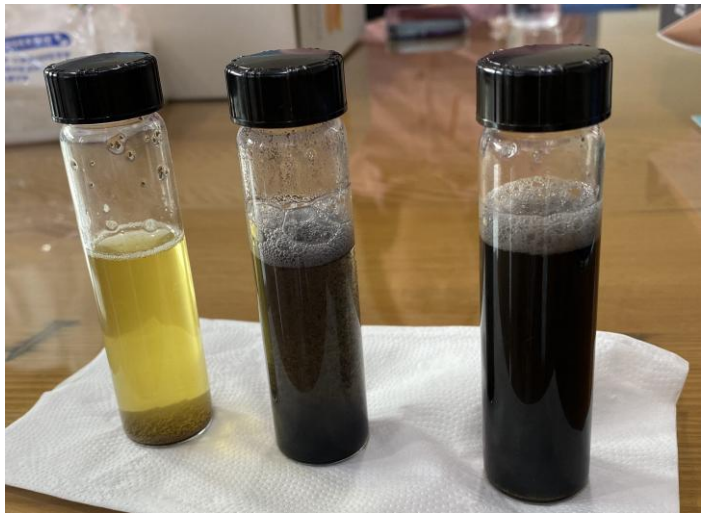
Biological Trace Element Research
https://doi.org/10.1007/s12011-023-03633-z

Organic Trace Minerals

Setback

1. Not all OTMs are created equal. Their chemical properties differ substantially: binding strength, solubility, retention rate
2. They are chemically complex, still oxidative, difficult for QC
3. Not a single solution
4. Exorbitant cost

Hot tea demo - oxidation



Added MPSC, ITM, OTM

3. Encapsulation "All-in-One"



Fe, Zn, Mn, Cu, Se, I



Special CHO as matrix

3. Encapsulating all TM: *All-in-One*

Select quality materials

↓

Formulate to premix

↓

Aqueous mixing with matrix

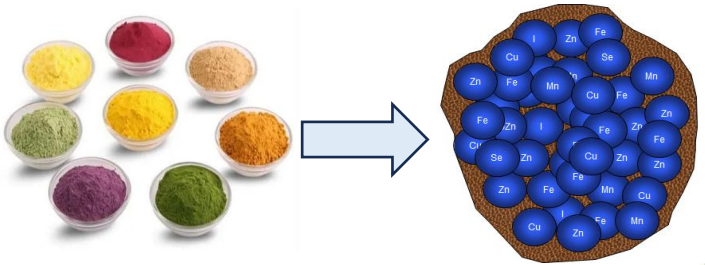
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Embedding (Reactions)

↓

Micro-pelleting & granulation

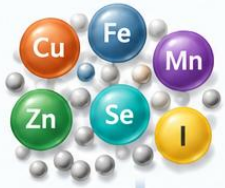




1. Encapsulation Process

Minerals are embedded in a starch matrix, forming protected granules (like coating a tablet).

1 Trace mineral premix



- Copper (Cu)
- Iron (Fe)
- Manganese (Mn)
- Zinc (Zn)
- Selenium (Se)
- Iodine (I)

2 Mix with starch matrix

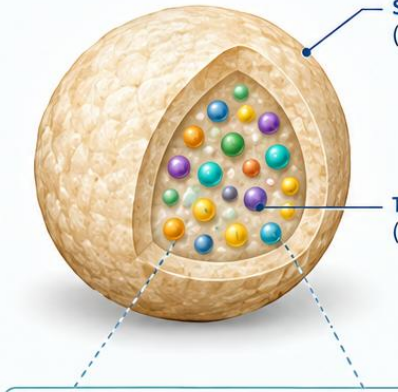


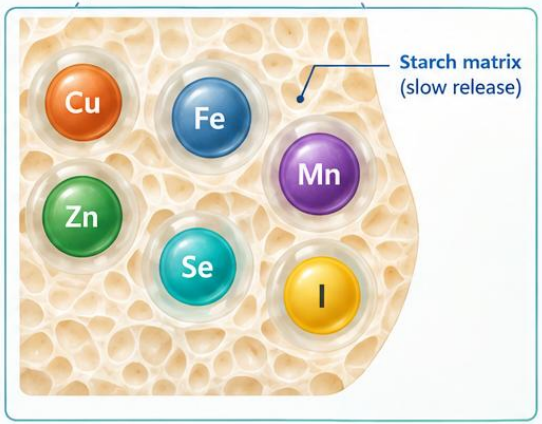
3 Granulation & drying



2. Cross-section of a Granule

Minerals are fully encapsulated in a starch matrix, forming a protective barrier.





3. Benefits

Protected minerals, better performance and less loss.

Stable
Protects from moisture, heat and oxidation

Slow release
Improves absorption in the intestine

Less interaction
Reduces mineral antagonism

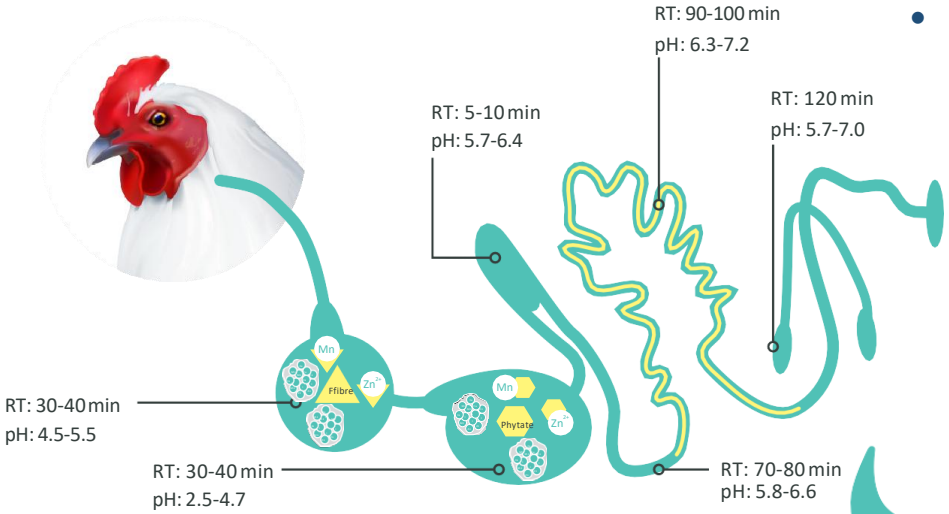
Better utilization
Higher bioavailability

Lower excretion
Supports animal health and sustainability

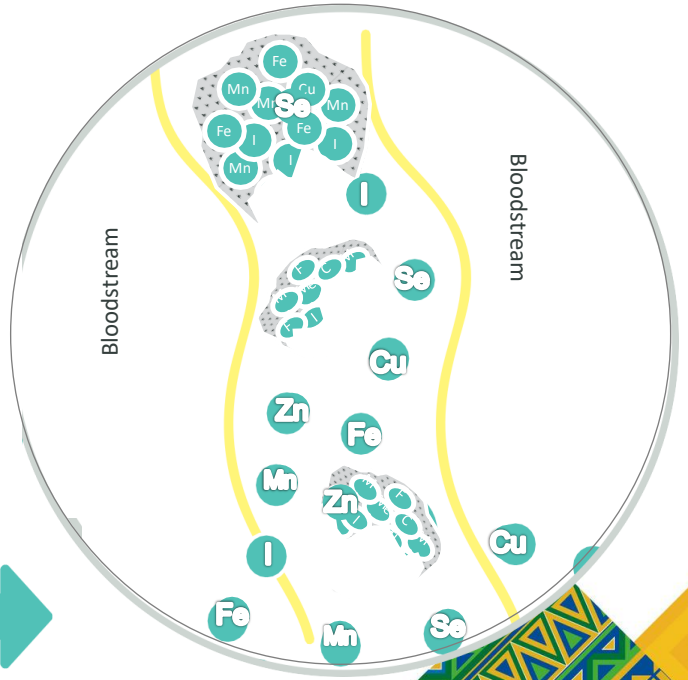
Starch-encapsulated trace minerals
= safer • more efficient • sustainable

Deliver to the GUT

- Plant-based matrix absorbs intestinal fluids, expands
- Slowly releases minerals on target site



Remains intact through the upper GIT



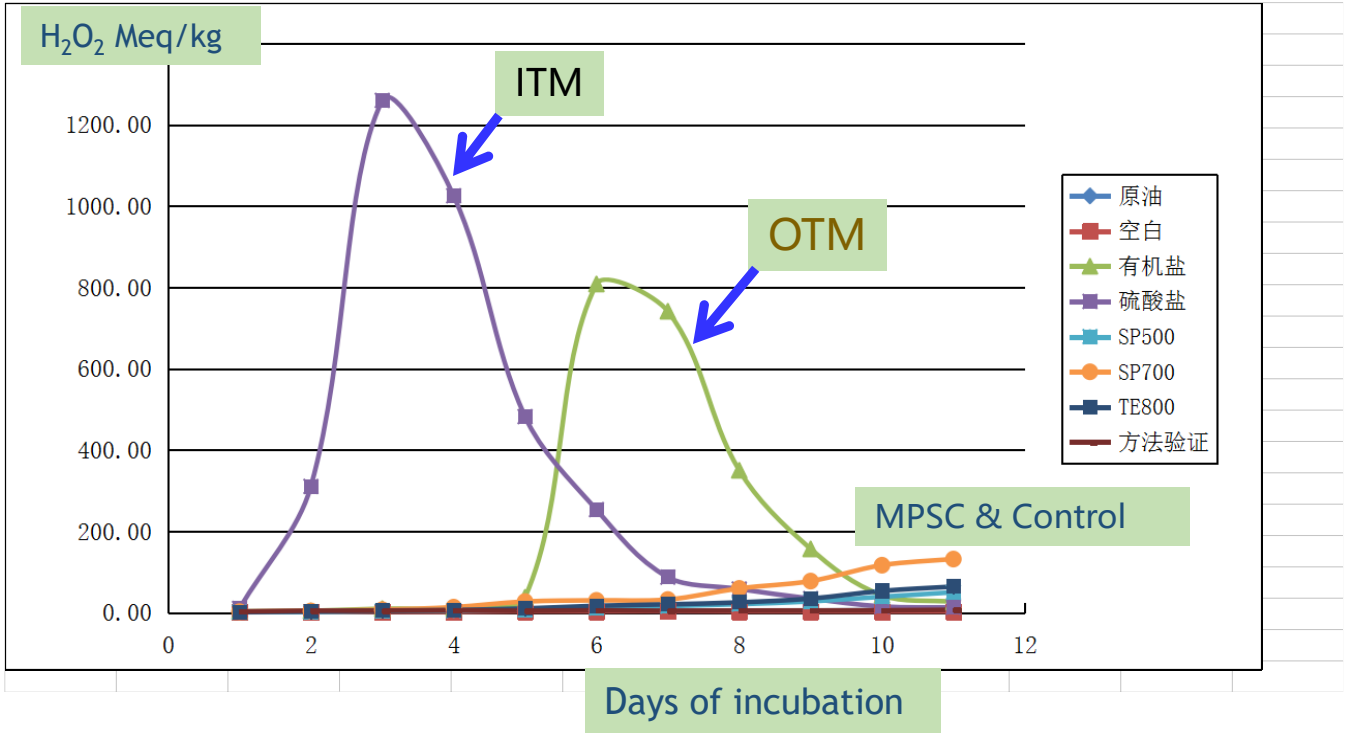
In vitro Test: Trace Minerals Oxidize Soy Oil

Test: Add edible oil in flask. A. Control (no trace minerals); B, C, D encapsulated premixes (MPSC); E. inorganic minerals. Incubate at 40 ^{o}C, measure H_2O_2 change.



Trace Minerals React with Soy Oil

Test: 150 g corn flour + 50 oil + Minerals, incubated at 40 °C, RH 75%, extract the oil and measure H_2O_2



Results: PUFA losses: 7 d for ITM and 10 D for OTM; Little changes in MPSC

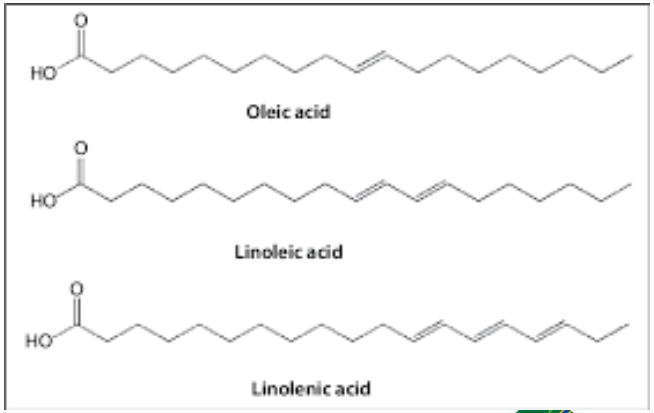
Mass Spectrometry - PUFA oxidation

Poly-Unsaturated Fatty Acids (PUFA) are essential nutrients

Fatty acid, %	Control	ITM	MPSC
Linolenic 亚麻酸	7.99	0.00	8.96
Linoleic 亚油酸	51.29	18.58	56.65
Palmitic 棕榈酸	12.19	24.32	8.69
Oleic 油酸	24.39	48.08	23.06
Stearic 硬脂酸	4.14	9.02	2.65

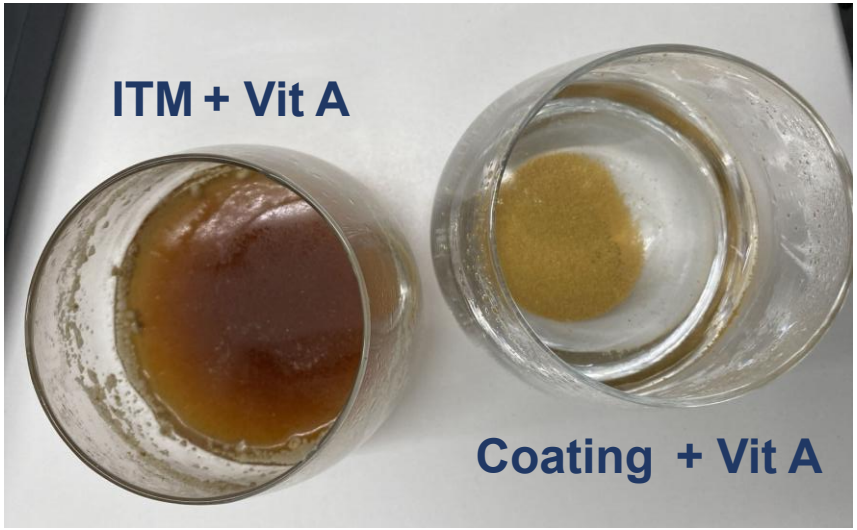
Results: After 5 d, ITM destroyed 100% linolenic acid, while coated premix showed full protection.





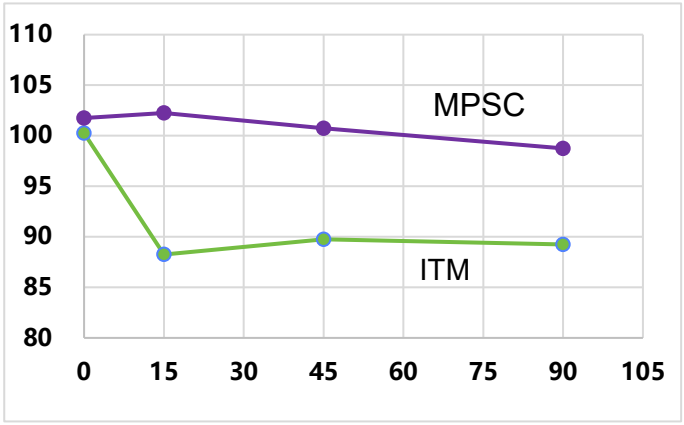
Trace minerals – Hidden Thief

1. Trace minerals leads to loss of dietary nutritional value by ca. 5%
2. Leads to over-supply of vitamins and enzymes
3. Shorten shelf-life of premix and feeds, esp. under tropical climate
4. Gut health problem and wet dropping

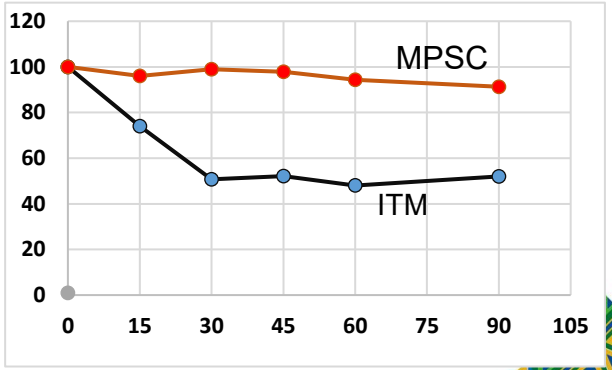


We can coat all trace minerals into a single granule, to prevent oxidizing fats, vitamins, phytase and gut epithelium

Phytase retention %



Canthaxanthin %



Days of storage at 25° C

In vitro test: Feed Rancidity

Pelleted feeds	Days to smell rancidity
Control: Basal diet of growing pigs, no added trace minerals, with 3% soy oil	**23**
Inorganic minerals, low Cu (Cu 20; Fe 120; Mn 25; Zn 120)	**13**
Inorganic minerals, high Cu (Cu 150; Fe 150; Mn 45; Zn 150)	**7**
+ OTM (Cu 10, Fe 50, Mn 20, Zn 60)	**9**
+ % ITM + 70% OTM (Low Cu formulation)	**7**
+ MPSC (Cu 66; Fe 70; Mn 20; Zn 60)	**22**

Note: Test in Aug., temp. 32-38.5°C

Broiler study: Form of minerals on growth of the first 10 days (The Netherlands)

	Wt gain, g	F. intake, g	FCR	Mortality, %
ITM	235	249	1.062 [b]	0.0
OTM	238	250	1.050 [b]	2.8
MPSC	248	256	1.034 [a]	0.0

Remiraz at al., 2023

Results: MPSC, ITM and OTM on broiler 37 d performance (The Netherlands)

0-37 days	WG (kg)	FI (kg)	FCR	Mortality (%)	EBI
ITM	2.463	3.602	1.463	5.0	432
OTM	2.515	3.659	1.456	8.5	428
MPSC-250 g	2.487	3.582	1.441	3.8	450
MPSC-375 g	2.530	3.654	1.445	4.2	453

Remiraz at al., 2023

No significant differences were observed for all treatments
EBI = European Broiler Index = (BWG (g/chick/day)*viability%)/(FCR*10)

Broiler Study 3: MPSC, ITM, OTM on broiler performance (MPSC usage 350/250/200 g/mt)

	ITM	OTM	MPSC
Fe	40	20 ($FeSO_4$ H_2O)	49/35/28
Cu	15	5 (Cu-Met)	4.55/3.25/2.60
Mn	100	40 (Mn-Met)	42/30/24
Zn	100	30 (Zn-Met)	42/30/24
Se	0.3	0.3 (Se-Met)	0.28/0.20/0.16
I	1.0	1.25 (KI)	0.28/0.20/0.16

Ross 308, male, 3 trts x 10 pen rep., 30 birds/rep., total 900 birds
MPSC usage: 350/250/200 g/mt, reps. for Starter / Grower / Finisher

Yuwares et al., 2021, Thailand

Efficacy of coated trace mineral complex MPSC on broiler performance (saves 70% Zn, Mn and 80% Cu)

	ITM	OTM	MPSC
Final wt, g/b	2649.5	2647.3	2652.5
Feed intake, g	3607.7	3582.6	3595.2
FCR	1.386	1.377	1.379
Mortality, %	1.702	2.368	1.357
Perf. Index	528	527	533

- Excellent performance, no differences among treatments, and Fe, Cu, Mn and Zn levels in the liver, confirmed descending supplementation of trace minerals
- MPSC usage: 0.844 g/bird, or 235 g/mt feed on average

(Dr. Yuwares et al. 2021, JAPR)

Trace minerals: Performance vs. Intake

	42d wt, kg	Feed intake, kg	FCR	Mortality, %
ITM	3209	4.92	1.556	0
MPSC	3247	4.96	1.550	1.0

Intake, mg/bird	Cu	Fe	Mn	Zn	Total
ITM	73.8	196.8	492.0	492.0	**1254.6**
MPSC	17.7	59.1	98.5	88.7	**264.1**

1. Ross 308, male, 2 trts x 12 pen rep., 10 birds/pen, total 240 birds
2. MPSC doses: 350/300/250/200 g/mt, reps. for Starter / Grower / Finisher
3. MPSC at descending doses saved 79% trace minerals, Cu 76% Zn 82%

Cu & Zn: Intake and Excretion

	Cu， mg/kg			Zn, mg/kg		
	Added	Total	Fecal	Added	Total	Fecal
ITM	15	23.1	92.7	100	157.5	376.2
MinCo®	2.5	11.1	**33.0**	15.6	54.2	**141.7**

1. Using MPSC to supply Cu & Zn, can reduce usage by 85%
2. Broiler achieved expected performance, carcass quality, no differences of minerals in liver and muscle
3. Levels Cu & Zn in feces dropped by 2/3

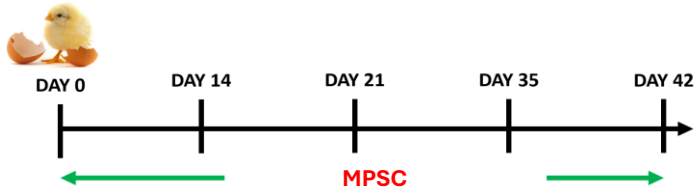
Recommendation, mg/kg

Phase	Days	Cu	Fe	Mn	Zn
Starter	1 - 10	3.0	18.0	15.0	19.0
Grower 1	11 - 24	2.5	15.0	12.5	15.6
Grower 2	25 – 34	2.0	12.0	10.0	12.5
Finisher	35 - 42	1.5	9.0	7.5	9.4

MPSC doses: 350/300/250/200 g/mt, reps. for Starter / Grower 1, 2/ Finisher
Save 79% trace minerals, Cu 76% and Zn 82%

Field results – 408,000 broiler - A Thai Integrator

- ✓ A total of 408,000 Cobb 500 birds were divided into 2 treatments: Control vs. MPSC, from 0-42 days of age
- ✓ Control diet followed the farm's existing program (ITM + 1 OTM), MPSC scheme was 350/300/250/200 g/mt resp. for the 4 growing phases, as the right chart
- ✓ Below results were obtained from its slaughter-house
- ✓ Time: June – July 2024



MPSC (pre-starter 350ppm), (starter 300ppm), (grower 250ppm), (finisher 200ppm)

	End wt, kg	ADG, g	FCR	Performance index
Control	2.65[b]	60.4 [b]	1.85	298 [b]
MPSC	2.71[a]	61.8 [a]	1.77	317 [a]
P value	0.018	0.020	0.212	0.049

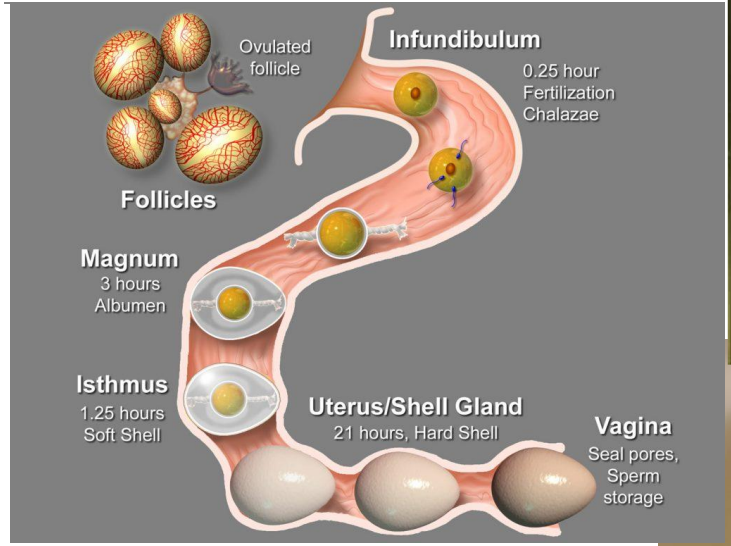
Conclusion:

1. Using MPSC to replace standard trace minerals sheme (ITM+OTM), MPSC usage was 1.239 g/bird or 242.5 g/mt feed
2. Cost of producing 1 kg live bird saved Baht 1.26 (25.88 vs. 24.62), resulting in extra revenue B 3.76 (U$0.1164) per bird, at the prevailing farm selling price B40/kg.

Aged laying hen – Oxidative Stress in Ovary & Oviducts

- ✓ Health status, age, oxidative stress



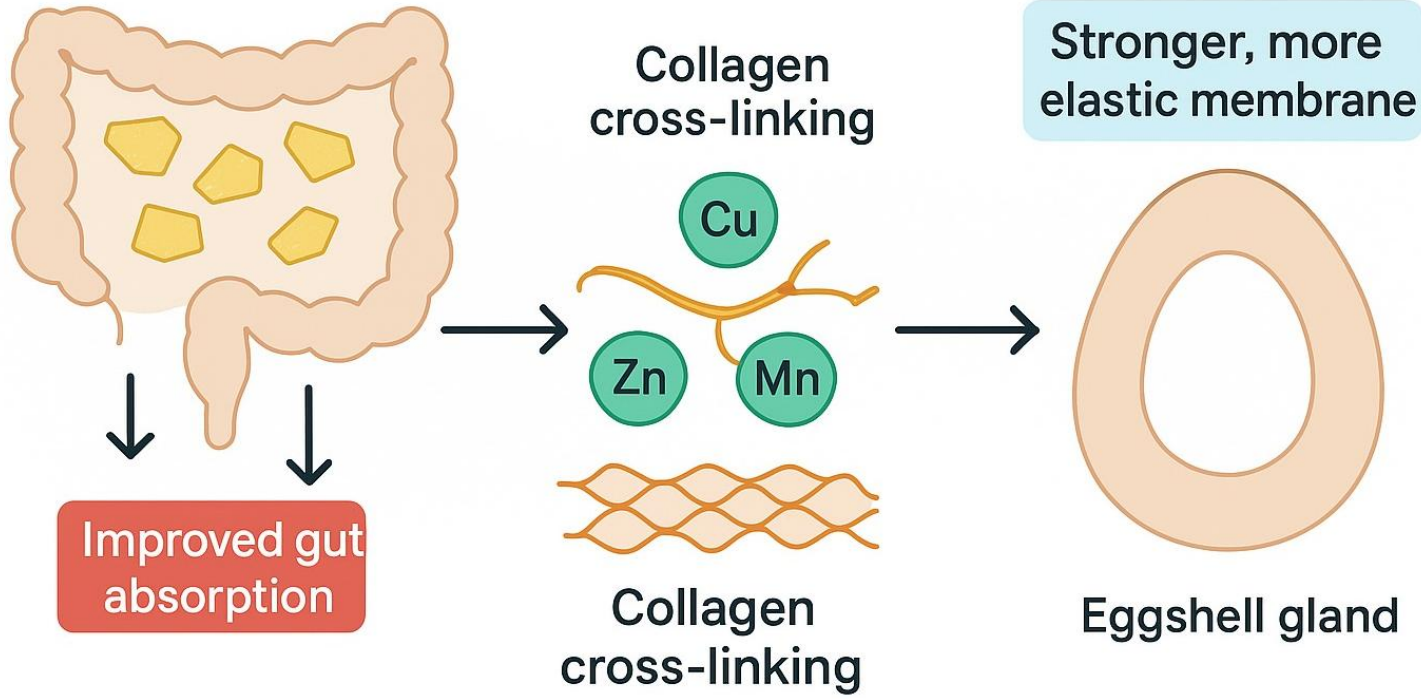






Watery albumen: from oxidative stress of ovary and oviducts

Slow-release trace minerals improve eggshell membrane formation and elasticity



Layer: Study in Malaysia (2022)

1. A field trial was conducted in a commercial layer farm, with 2 flocks of 64 weeks of age. Its trace mineral program was Inorganic + Organic, replaced by MPSC (400 g/mt).
2. Test period: Prior observation Jul 18-30; Using MPSC Jul 31 – Sept 18.
3. Egg samples were taken after 3 weeks of MPSC application, measured by an independent lab using standard egg tester.

	Grade AA, %	Yolk colour	Albumen height, mm	Haugh U. mm	Shell strength, kgF	Shell thickness, mm	Dirty * egg, %	Egg wt, g*
Before	59	11.7 [b]	5.5 [b]	70.7 [b]	4.1	0.41	9.2 [a]	60.9
MPSC	100	13.2 [a]	6.8 [a]	82.1 [a]	4.0	0.40	7.9 [b]	62.5

Results: Using MPSC significantly (P<0.05) improved yolk colour, albumen height, haugh Unit, and reduced % dirty eggs. All eggs from hens fed on MPSC are Grade AA. No changes were observed on eggshell strength and thickness. * Observation at farm.

MPSC Poultry Spec, g/kg

Cu	Fe	Mn	Zn	Se	I	Co
13	140	120	120	0.8	0.8	0.6

Broiler	g/mt feed
Starter	350
Grower	250
Finisher	200

Layer	g/mt feed
Pullets	300
Peak lay	400
Aged hen	450-500

Broiler feed, average usage: 235 g/mt feed

Conclusion

1. It is a global problem to define exact requirements of trace minerals, leading to over-supply in the industry

   Impact 1: Oxidation, damages nutrients and gut health, can be 5%

   Impact 2: High excretion, esp Cu & Zn

2. Outlook: Give up ITM, use either high quality OTM or MSPC, as stable and standard form of trace minerals

   Use phytase + MPSC + add based on growth age (e.g. 1.3/1.0/0.75)

We shall be able to reduce Cu & Zn by at least 50%